BT301 Mid Term Solve
Subjective

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Mid BT301 Solved

1. Chargaff Rules

1. Base composition of DNA varies from one species to another.
2. The DNA isolated from different tissues of the same species has the same base composition.
3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.
4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C).

2. R-M system mechanism (5)

After the discovery of large number of Type II endonucleases, scientists proposed that there must be a naming mechanism which we can call as the nomenclature of the restriction endonucleases.

- A suitable system was proposed by Smith and Nathans (1973)
- The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet-
- *E. coli* = Eco
- *H. influenzae*= Hin
- Strain identification is written as *Eco*K
- In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of *H. influenzae*
- *HindI, HindII, HindIII*etc
- All restriction enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of *H. influenzae*
- R. *HindIII*or M. *HindIII*

3. Three major and minor groove , number of nucleotide in them (3)

The **major groove** occurs where the backbones are far apart, the **minor groove** occurs where they are close together. The **grooves** twist around the molecule on opposite sides. Certain proteins bind o **DNA** to alter its structure or to regulate transcription (copying **DNA** to RNA) or replication (copying **DNA** to **DNA**).

It is 23.7 Å wide and extends 34 Å per 10 bp of sequence. The double helix makes one complete turn about its axis every 10.4-10.5 base pairs in solution.

4. Euchromatic & Heterochromatin

Chromatin is a complex of macromolecules found in cells, consisting of DNA, protein, and RNA. The primary functions of chromatin are 1) to package DNA into a more compact, denser shape, 2) to reinforce the DNA macromolecule to allow mitosis, 3) to prevent DNA damage, and 4) to control gene expression and DNA replication. . That DNA which codes genes that are actively transcribed ("turned on") is more loosely packaged and associated with RNA polymerases (referred to as euchromatin) while that DNA which codes inactive genes ("turned off") is more condensed and associated with structural proteins (heterochromatin)

- Highly condensed chromatin is known as **heterochromatin**.
- The more extended form is known as **euchromatin.**

5. Natural plasmid

- Plasmids which were not constructed *in vitro* for the sole purpose of cloning are called natural plasmids
- Col E1 is a natural occurring plasmid
- RSF2124 is a derivative of Col E1
- pSC101 is natural plasmid from *Salmonella panama*

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6. Principle of western blotting?

Western blotting

- It involves the transfer of electrophoreses protein bands from a gel on to a membrane
- The bounded proteins are then available for analysis
- Most commonly, antibodies are used to detect specific antigens

7. Define proteomics?

Proteome and Proteomics:

- The complete set of proteins found in each cell is known as the proteome and the study of proteome is called as proteomics.

8. Define Genome?

Genome and Genomics:

- The complete set of DNA found in each cell is known as the genome and study is called as genomics.

9. Palindromic Sequence?

A **palindromic sequence** is a **sequence** made up of nucleic acids within double helix of DNA and/or RNA that is the same when read from 5' to 3' on one strand and 5' to 3' on the other, complementary, strand. It is also known as a **palindrome** or an inverted-reverse**sequence**.

10. Name of any five amino acids?

- Alanine, Ala, A. Isoleucine, Ile, I. ...
- Valine, Val, V.
- Phenylalanine, Phe, F. Tryptophan, Trp, W. ...
- Asparagine, Asn, N. Cysteine, Cys, C. ...
- Threonine, Thr, T.

11. Work of Franklin & Wilkins?

Work of Franklin & Wilkins (1950s):

- Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibres.
- They showed that DNA produces a characteristic x-ray diffraction pattern.
- From this pattern, they made two important findings:-

1. DNA molecules are helical.

2. The helices have two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å.

12. What is pseudo gene?

Pseudo-Genes

• Nonfunctional copies of genes

• Formed by duplication of ancestral gene, or reverse transcription and integration.

13. Standard amino acids name?

All the proteins are made up of twenty different types of amino acids. So these amino acids are called standard amino acids.

14. Write about electroporation role?

Electroporation

- A rapid and simple technique for introducing cloned genes into a wide variety of microbial, plant and animal cell, including *E. coli,* is electroporation
- In electroporation, electric field is applied to cells in order to increase permeability of the cell

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membrane to take DNA

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15. Write different structures of proteins.

The **different** levels of **protein structure** are known as primary, secondary, tertiary, and quaternary **structure**. The primary **structure** is the sequence of amino acids that make up a polypeptide chain. 20 **different** amino acids are found in **proteins**.

Primary: The primary structure of a protein is the level of protein structure which refers to the specific sequence of amino acids [1]. When two amino acids are in such a position that the carboxyl groups of each amino acid are adjacent to each other, they can be combined by undergoing a dehydration reaction which results in the formation of a peptide bond

Secondary: The amino acid sequence of a polypeptide, together with the laws of chemistry and physics, cause a polypeptide to fold into a more compact structure. Amino acids can rotate around bonds within a protein.

Characteristics of the Secondary Structures:

1. α-helix

2. **β-sheet**:

Tertiary: As the secondary structure becomes established due to the primary structure, a polypeptide folds and refolds upon itself to assume a complex three-dimensional shape called the protein tertiary structure.

Quaternary: While all proteins contain primary, secondary and tertiary structures, quaternary structures are reserved for proteins composed of two or more polypeptide chains [1]. Proteins that have quaternary structures contain more than one polypeptide and each adopt a tertiary structure and then assemble with each other via intermolecular interactions. The quaternary structure of a protein is the overall structure that is the result of the addition of these polypeptide subunits

16. Write one differential point between super coiled and relaxed loops?

Some of the loops are super-coiled, while some are relaxed, Relaxed loops are formed as a result of a nick introduced into super-coiled loops by a cellular DNase, If a super-coiled DNA molecule receives a nick, the strain of underwinding is immediately removed, and all the super-coiling is lost, Studies confirm that continued nuclease treatment increases number of relaxed loops. The bacterial DNA is arranged in super-coiled loops that are fastened to a central protein matrix, so that each loop is topologically independent from all the others. So a nick that causes one super-coiled loop to relax would have no effect on other super-coiled loops. The super-coiled loops are dynamic structures which change during cell growth & division. An *E. coli* chromosome is estimated to have about 400 super-coiled loops. Each loop has an average length of about 10-20 kbp. The DNA compaction in a bacterial cell is contributed by super-coiling of loops, macromolecular crowding and DNA-binding proteins.

17. Green Biotechnology?

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones. An additional gene was inserted into the normal genome of this corn that triggers the production of proteins that then protect the plant from harmful insects. This reduces the need for pesticides and increases crop yields

Green BiotechnologyExample:

Bt corn from Syngenta
Bt cotton from Monsanto
sprigene control.

18. Griffith Experiment?

Griffith's experiment was an experiment done in 1928 by Frederick Griffith. It was one of the first experiments showing that bacteria can get DNA through a process called transformation. Griffith used two strains of Streptococcus pneumoniae. These bacteria infect mice,Griffith's favorite animals.

19. Restriction modification?

Host-controlled restriction/ modification

- Restriction system allow bacteria to monitor the origin of incoming DNA
- When the incoming DNA is a bacteriophage genome, the effect is to reduce the efficiency of plating (Efficiency of plating means the number of plaques that are produced during the lysis) i.e. to reduce the number of plaques in plating test
- The phenomenon of restriction and modification were well illustrated and studied by the behavior of phage ʎ on two *E. coli* strains

E. coli C and *E. coli* K

Host-controlled restriction/modification

- Restrictive host must, of course protects its own DNA from restriction endonucleases by modification
- Modification involves methylation of certain bases that constitute recognition sites

Types of restriction and modification (R-M) system:

- At least four R-M systems are known
- Type I
- Type II
- Type III
- Type IIs

hairpin like structure of RNA

- RNA has no any regular secondary structure that serves as a reference point.The three-dimensional structures of many RNAs are complex and unique.
- Breaks in the helix caused by mismatched or unmatched bases in one or both strands are common and result in bulges or internal loops.
- Hairpin loops form between nearby self-complementary sequences.

20. Role of primase in antibiotic

Primase produces RNA molecules, the enzyme is a type of RNA polymerase.**Primase functions** by synthesizing short RNA sequences that are complementary to a single-stranded piece of DNA, which serves as its template. It is critical that primers are synthesized by **primase** before DNA replication can occur.

21. Pregnancy test?

A papyrus described a test in which a woman who might be pregnant could urinate on wheat and barley seeds over the course of several days: –If the barley grows, it means a male child. If the wheat grows, it means a female child. If both do not grow, she will not bear at all.l elevated levels of estrogens in pregnant women's urine may have been the key to its success.

22. What is nuclein?

Miescher isolated various phosphate-rich chemicals, which he called **nuclein** (now nucleic acids), from the nuclei of white blood cells in 1869. The term used by Friedrich Miescher to describe the nuclear material he discovered in 1869, which today is known as DNA.

23. What is the central dogma of molecular biology?

The conversion of DNA to mRNA is transcription, and the conversion of mRNA to the protein is translation. Besides messenger RNA (mRNA) , there is also ribosomal RNA (rRNA) and transfer RNA (tRNA) , but here we are concerned with mRNA. A protein is simply a large peptide, and can also be called a polypeptide.

24. Relaxed plasmid?

A plasmid that replicatesindependently of the main bacterial chromosome and is present in 10-500 copies per cell.

25. Benefit of genomic in madicine research?

Genomes Research – Molecular Medicine?

- Improve diagnosis of disease
- Detect genetic predispositions to disease (cancer, diabetes etc.)
- Create drugs based on molecular information
- Use gene therapy and control systems as drugs

26. Biotechnology and types?

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize,break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

(i) White Biotechnology:

Development of processes and microorganisms for Industrial processes.

White BiotechnologyExample:

Enzyme production

(ii) Red Biotechnology:

It is concerned with the discovery and development of innovative drugs and treatments.

(iii) Green Biotechnology:

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones.

Green BiotechnologyExample:

Bt corn from Syngenta
Bt cotton from Monsanto

27. What factor affect the DNA mobility in electrophoresis gel?

- Concentration of gel either high concentration or lower concentration.
- Ionic strength of buffer.
- Voltage
- Ethidium bromide (A chemical used as a stain to dye gel)

28. Major minor groove its length and how many nucleotides are present?

29. What type of cell can be transformed using liposomes?

Animal cells, protoplasts (without cell wall) of yeast and plant are susceptible to transformation by liposomes.

30. Differentiate b/w protomers, oligomers and multimers?

- A multi-subunit protein is also referred to as a multimer.
- A multimer with just a few subunits is called as **oligomer** and a single subunit or a group of subunits, is called a **protomer.**

31. Importance of Ori region in plasmid?

Host range of plasmids

- When plasmids are present in microbes, whether they have narrow host range or wide host range.
- Plasmids encode only a few of the proteins required for their own replication
- Host range of plasmid is determined by *ori* region (Point where DNA replication mainly starts)
- Plasmids whose *ori* derived from Col E1 have restricted host range
- RP4 and RSF1010 and many plasmids from *Staphylococcus aureus* have broad host range.

32. Retrotranspos in humans?

- Retrotransposons constitute over 40% of the human genome and consist of several millions of family members.
- They play important roles in shaping the structure/evolution of the genome.

33. What are 3 units for the formation of nucleotide?

- A molecule containing all three components (Nitrogenous base, sugar molecule and phosphoric acid) is called a nucleotide.
- While a molecule without the phosphate group is called a nucleoside.
- During 1960s phage biologist elucidated the phenomenon of restriction and modification

34. Side effect of ethidium bromide?

Effect of ethidium bromide on supercoiling of DNA

Some chemical agents can also influence the super coiling of DNA depending upon its concentration that is used. One of the chemical agents is Ethidium bromide.

- DNA bands can be visualized by soaking the gel in a solution of ethidium bromide
- As the amount of intercalating agent i.e. ethidium bromide increases, the double helix untwists until the open form of circular DNA is produced
- Ethidium bromide may be a mutagen, a carcinogen, or a teratogen
  Health Effects. Because ethidium bromide can bind with DNA, it is highly **toxic** as a mutagen. It may potentially cause carcinogenic or **teratogenic effects**. Exposure routes of ethidium bromide are **inhalation**, **ingestion**, and skin absorption.

35. Cole 1?

- ColE1 is a naturally occurring plasmid of *E. coli*. Its replication is controlled independently of the replication of the host chromosome.

36. Selfish DNA?

Selfish DNA is a term for sequences of DNA that *sensu stricto* have two distinct properties:
the DNA sequence spreads by forming additional copies of itself within the genome; and

- it makes no specific contribution to the reproductive success of its host organism. (It might or might not have significant deleterious effects.)

Selfish DNA is non-transcribed, non-coding, and contributes nothing to the well-being of the organism; in most cases it is selectively neutral.

37. Describe bonding between parts of a nucleotide?

The links between the nucleotides are called phosphodiester bonds.

38. Briefly describe Type-III R-M?

Type III restriction-modification enzymes: a historical perspective. ... Based on their molecular structure, sequence recognition, cleavage position and cofactor requirements, restriction-modification (R-M) systems are classified into four group

39. RAPD?

RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random.

40. Restriction modification phenomena?

Restriction modification system. The **restriction modification** system (RM system) is found in bacteria and other prokaryotic organisms, and provides a defense against foreign DNA, such as that borne by bacteriophages. It was first discovered by Salvatore Luria and Mary Human in 1952 and 1953.

41. X-ray diffraction?

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Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibers. They showed that DNA produces a characteristic x-ray diffraction pattern.

42. Gene 2?

A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

43. Ori region?

Plasmids encode only a few of the proteins required for their own replication § Host range of plasmid is determined by ori region § Plasmids whose ori derived from Col E1 have restricted host range.

44. Major technique use for genome characterization?

A whole range of different techniques can be used to detect polymorphisms at the DNA level. In fact the seemingly bewildering array of possible approaches is among the first problems faced by newcomers considering the application of these techniques to their own system. In reality, however, this wide array falls into three broad categories with respect to basic strategy: (A) Non-PCR based approaches; (B) PCR Arbitrary priming; and (C) Targeted-PCR and sequencing.

45. Reverse transcriptase?

A **reverse transcriptase** (RT) is an enzyme used to generate complementary DNA (cDNA) from an RNA template, a process termed**reverse transcription**.

46. Nomenclature for restriction enzyme?

Restriction enzymes are named based on the organism in which they were discovered. For example, the **enzyme** Hind III was isolated from Haemophilus influenzae, strain Rd. The first three letters of the name are italicized because they abbreviate the genus and species names of the organism.Aug 13, 2017

47. Benefits of genomic research?

Genome sequencing allows scientists to isolate the DNA of an individual person and identify different codes. Then, based on scientific studies of patients with diseases like **cancer** or heart disease, scientists can pinpoint the common DNA letter codes--acting as a predictor.Oct 4, 2011

48. Function of retrotransposons in human genome?

In wheat, about 90% of **the genome** consists of repeated sequences and 68% of transposable elements. In mammals, almost half **the genome**(45% to **48**%) is transposons or remnants of transposons. Around 42% of **the human genome** is made up of **retrotransposons**, while **DNA**transposons account for about 2–3%.

49. Justify influence virus in one line?

Influenza viruses bind through **hemagglutinin** onto sialic acid sugars on the surfaces of epithelial cells, typically in the nose, throat, and lungs of mammals, and intestines of birds (Stage 1 in infection figure). After the **hemagglutinin** is cleaved by a protease, the cell imports the virus by endocytosis.

50. RNA types ?

There are 4 types of RNA, each encoded by its own type of gene:

- mRNA - Messenger RNA: Encodes amino acid sequence of a polypeptide.
- tRNA - Transfer RNA: Brings amino acids to ribosomes during translation.
- rRNA - Ribosomal RNA: With ribosomal proteins, makes up the ribosomes, the organelles that translate the mRNA

51. Northern blotting?

The **northern blot**, or RNA **blot**, is a technique used in molecular biology research to study gene expression by detection of RNA (or isolated mRNA) in a sample.

52. What is meant by Nicking of DNA?

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A **nicking** enzyme (or **nicking** endonuclease) is an enzyme that cuts one strand of a double-stranded **DNA** at a specific recognition nucleotide sequences known as a restriction site. Such enzymes hydrolyse (cut) only one strand of the **DNA** duplex, to produce **DNA** molecules that are **-nicked**l , rather than cleaved.

53. What is difference between?

A protomer is the structural unit of an oligomeric protein. It is the smallest unit composed of at least two different protein chains that form a larger hetero-oligomer by association of two or more copies of this unit. The term multimer (/ˈmʌltɪmər/) (multi-, "more than one" + -mer, "part") is used with a meaning similar to that of oligomer in the context of proteins (although technical restrictions of word sense may exist) It can also refer to a protein complex made of two or more subunits.

54. Note on Restriction-Modification principle?

In order to prevent destruction of its own DNA by the restriction enzymes, methyl groups are added. These modifications must not interfere with the DNA base-pairing, and therefore, usually only a few**specific** bases are modified on each strand.

55. What was pregnancy test performed to test gender in ancient period?

56. What is the role of transposes in human genome?

Transposable elements (TEs) are mobile repetitive sequences that make up large fractions of mammalian **genomes**, including at least 45% of the **human genome**(Lander et al. 2001), 37.5% of the mouse **genome** (Waterston et al. 2002), and 41% of the dog **genome** (Lindblad-Toh et al. 2005).

57. What was the Frederick grifth experiment about genetic material?

Experiment: **Griffith** injected both S and R strains to mice. .....The R strain inherited some 'transforming principle' from the heat-killed S strain bacteria which made them virulent. And he assumed this transforming principle as **genetic material**.

58. What is the role of electriporation technique?

Electroporation. **Electroporation**, or electropermeabilization, is a microbiology **technique** in which an electrical field is applied to cells in order to increase the permeability of the cell membrane, allowing chemicals, drugs, or DNA to be introduced into the cell (also called electrotransfer).

59. What is the main components consisting the RNA molecule?

RNA stands for ribonucleic acid and like DNA, RNA nucleotides contain three components:

- A Nitrogenous **Base**.
- A Five-Carbon Sugar.
- A Phosphate Group

60. Define autoradiography?

Autoradiography is the use of X-ray (or occasionally photographic) film to detect radioactive materials. It produces a permanent record of the positions and relative intensities of radiolabeled bands in a gel or blot. Typically, biomolecules are labeled with ^{32}P or ^{35}S, and detected by overnight film exposure.

61. Define traditional biotechnology?

Traditional biotechnology refers to ancient ways of using living organisms to make new products or modify existing ones. An example of **traditional biotechnology** is what human beings have been doing for centuries: cloning plants.

62. It is concerned with the discovery and development of innovative drugs and treatments. ?

Red biotechnology(Bio-pharmaceutical) is a process that utilizes organisms to improve health care and help the body to fight diseases. It is a branch of modern biotechnology which is utilized in the field of medicine.

63. It is used in lyses (protease K)?

The residual activity is sufficient to digest proteins, which usually contaminate nucleic acid preparations. Therefore, the digestion with **Proteinase K** for the purification of nucleic acids is usually performed in the presence of EDTA (inhibition of calcium-dependent enzymes such as nucleases).

64. LTR consists of …. Bp?

Retrotransposons are **made up** of short interspersed nuclear elements (SINES), long interspersed nuclear elements [LINES, also known as non-long terminal repeat (**LTR**) retrotransposons], **LTR** retrotransposons, and retroviruses.

65. Nucleotide differs due to change in???

Bases

66. The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water? PPT

Work of Watson & Crick

The nitrogenous bases of both strands are stacked inside the double helix lying perpendicular to the long axis of the helix.

67. How many bases DNA and RNA have?

RNA is a polymer with a ribose and phosphate backbone and four different **bases**: adenine, guanine, cytosine, and uracil. The first three are the same as those found in **DNA**, but in **RNA** thymine is replaced by uracil as the **base** complementary to adenine.

68. Mrna transcribe the template into Protein?

During **transcription**, the enzyme **RNA** polymerase (green) uses DNA as a **template to** produce a pre-**mRNA** transcript (pink). The pre-**mRNA** is processed **to** form a mature**mRNA** molecule that can be translated **to** build the **protein**molecule (polypeptide) encoded by the original gene.

69. How many functional groups attach to alpha carbon including alcohol R group?

(CHO) having a carbonyl **functional group attached**to one hydrocarbon radical and a hydrogen atom. carboxylic acid: Any of a class of organic compounds containing a carboxyl **functional group**—a **carbon**with a double bond to an oxygen and a single bond to another oxygen, which is in turn bonded to a hydrogen.

70. The bacterial chromosome is a very long (up to 1mm?

71. PCR amplification is used to amplify?

Polymerase chain reaction (**PCR**) is a **method** widely **used**in molecular biology to make many copies of a specific DNA segment. Using **PCR**, a single copy (or more) of a DNA sequence is exponentially **amplified** to generate thousands to millions of more copies of that particular DNA segment.

72. Yeast genome was completely sequenced by?

DNA was first **sequenced** in 1977. The first free-living organism to have its **genome completely sequenced** was the bacterium Haemophilus influenzae, in 1995.